Homework 19.10.26 – Exercises 20, 22–26

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Exercise 20

Given that f has L-Lipschitz gradient, i.e.,

$$\|\nabla f(x) - \nabla f(y)\| \le L\,\|x-y\| \qquad \forall x, y \in \mathbb{R}^d.$$

Show that

$$-\frac{L}{2}\,\|x-y\|^2 \le D_f(x,y) \le \frac{L}{2}\,\|x-y\|^2 \qquad \forall x, y \in \mathbb{R}^d.$$

Proof

It is enough to show that $|D_f(x,y)| \le \frac{L}{2}\,\|x-y\|^2$. Let $z := y + t(x-y)$, so that $x - y = \frac{z-y}{t}$. Then

$$\begin{aligned}
|D_f(x,y)| &= \big|f(x) - f(y) - \langle \nabla f(y), x-y\rangle\big| \\
&= \left|\int_0^1 \big\langle \nabla f\big(y + t(x-y)\big), x-y\big\rangle\, dt - \langle \nabla f(y), x-y\rangle\right| \\
&= \left|\int_0^1 \langle \nabla f(z) - \nabla f(y), z-y\rangle \cdot \frac{1}{t}\, dt\right| \le \int_0^1 \frac{1}{t}\,\big|\langle \nabla f(z) - \nabla f(y), z-y\rangle\big|\, dt.
\end{aligned}$$

By Cauchy–Schwarz and then by the L-Lipschitz gradient assumption,

$$\begin{aligned}
|D_f(x,y)| &\le \int_0^1 \frac{1}{t}\,\|z-y\|\,\|\nabla f(z) - \nabla f(y)\|\, dt \le \int_0^1 \frac{L}{t}\,\|z-y\|^2\, dt = \int_0^1 \frac{L}{t}\,\langle z-y, z-y\rangle\, dt \\
&= \int_0^1 L\,\langle x-y, x-y\rangle\, t\, dt = \frac{L}{2}\,\|x-y\|^2,
\end{aligned}$$

where we used $z - y = t(x-y)$.

Exercise 22

Let $T_\gamma(x) := x - \gamma\nabla f(x)$. Show that if f has L-Lipschitz gradient, then for all $x \in \mathbb{R}^d$ and $\gamma \ge 0$

$$\gamma\left(1 - \frac{\gamma L}{2}\right)\|\nabla f(x)\|^2 \le f(x) - f\big(T_\gamma(x)\big) \le \gamma\left(1 + \frac{3\gamma L}{2}\right)\|\nabla f(x)\|^2.$$

Proof

Let $x_t := x - t\gamma\nabla f(x)$. From the Fundamental Theorem of Calculus we get

$$\begin{aligned}
f(x) - f\big(T_\gamma(x)\big) &= \int_0^1 \Big\langle \nabla f\Big(x - t\big(x - (x - \gamma\nabla f(x))\big)\Big), x - \big(x - \gamma\nabla f(x)\big)\Big\rangle\, dt \\
&= \int_0^1 \big\langle \nabla f\big(x - t\gamma\nabla f(x)\big), \gamma\nabla f(x)\big\rangle\, dt = \gamma\int_0^1 \langle \nabla f(x), \nabla f(x_t)\rangle\, dt.
\end{aligned}$$

Also we have

$$\langle \nabla f(x), \nabla f(x_t) \rangle = \|\nabla f(x)\|^2 + \langle \nabla f(x), \nabla f(x_t) - \nabla f(x) \rangle .$$

As a result,

$$f(x) - f\big(T_\gamma(x)\big) = \gamma \|\nabla f(x)\|^2 + \gamma \int_0^1 \langle \nabla f(x), \nabla f(x_t) - \nabla f(x) \rangle \, dt.$$

So if we prove that

$$-\frac{\gamma L}{2} \|\nabla f(x)\|^2 \le \int_0^1 \langle \nabla f(x), \nabla f(x_t) - \nabla f(x) \rangle \, dt \le \frac{3\gamma L}{2} \|\nabla f(x)\|^2 ,$$

then the statement follows. On the other hand, by Cauchy–Schwarz and the L-Lipschitz gradient,

$$\begin{aligned} \big|\langle \nabla f(x), \nabla f(x_t) - \nabla f(x) \rangle\big| &\le \|\nabla f(x)\| \cdot \|\nabla f(x_t) - \nabla f(x)\| \le L \|\nabla f(x)\| \cdot \|x_t - x\| \\ &= L \|\nabla f(x)\| \cdot \|t\gamma \nabla f(x)\| = L\gamma t \|\nabla f(x)\|^2 . \end{aligned}$$

As a result,

$$\left| \int_0^1 \langle \nabla f(x), \nabla f(x_t) - \nabla f(x) \rangle \, dt \right| \le L\gamma \|\nabla f(x)\|^2 \int_0^1 t \, dt = \frac{L\gamma}{2} \|\nabla f(x)\|^2 ,$$

which proves the statement, because $\frac{L\gamma}{2} \le \frac{3L\gamma}{2}$.

Exercise 23

Find an upper bound for $\left\|x^{k+1} - x^*\right\|^2$ using Proposition 20, i.e.: if $f\colon \mathbb{R}^d \to \mathbb{R}$, $f \in C^1(\mathbb{R}^d)$, is also μ-convex and L-smooth, then

$$\mu \|x - y\|^2 + \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2 \le \left(1 + \frac{\mu}{L}\right) \langle \nabla f(x) - \nabla f(y), x - y \rangle \qquad \forall x, y \in \mathbb{R}^d.$$

Proof

$$\begin{aligned} \left\|x^{k+1} - x^*\right\|^2 &= \left\|(x^k - x^*) - \gamma\big(\nabla f(x^k) - \nabla f(x^*)\big)\right\|^2 \\ &= \left\|x^k - x^*\right\|^2 + \gamma^2 \left\|\nabla f(x^k) - \nabla f(x^*)\right\|^2 - 2\gamma \left\langle \nabla f(x^k) - \nabla f(x^*), x^k - x^* \right\rangle . \end{aligned}$$

On the other hand, multiplying Proposition 20 by $\frac{L}{L+\mu}$,

$$\left\langle \nabla f(x^k) - \nabla f(x^*), x^k - x^* \right\rangle \ge \frac{L\mu}{L+\mu} \left\|x^k - x^*\right\|^2 + \frac{1}{L+\mu} \left\|\nabla f(x^k) - \nabla f(x^*)\right\|^2 .$$

Multiplying by -2γ:

$$-2\gamma \left\langle \nabla f(x^k) - \nabla f(x^*), x^k - x^* \right\rangle \le -\frac{2L\mu\gamma}{L+\mu} \left\|x^k - x^*\right\|^2 - \frac{2\gamma}{L+\mu} \left\|\nabla f(x^k) - \nabla f(x^*)\right\|^2 .$$

As a result,

$$\left\|x^{k+1} - x^*\right\|^2 \le \left(1 - \frac{2L\mu\gamma}{L+\mu}\right) \left\|x^k - x^*\right\|^2 - \gamma \left(\frac{2}{L+\mu} - \gamma\right) \left\|\nabla f(x^k) - \nabla f(x^*)\right\|^2 .$$

In particular, for $0 \le \gamma \le \frac{2}{L+\mu}$ the last term is non-positive, and we get

$$\left\|x^{k+1} - x^*\right\|^2 \le \left(1 - \frac{2L\mu\gamma}{L+\mu}\right) \left\|x^k - x^*\right\|^2 .$$

Note that

$$1 - \frac{2L\mu\gamma}{L+\mu} \le 1 - \mu\gamma, \qquad \text{because} \qquad \frac{2L}{L+\mu} \ge 1.$$

Exercise 24

For all $a, b \in \mathbb{R}^d$ and $t > 0$ the following inequalities hold:

$$\begin{aligned} &1) \quad \langle a, b\rangle \leq \frac{\|a\|^2}{2t} + \frac{t\,\|b\|^2}{2}, \\ &2) \quad \|a+b\|^2 \leq 2\,\|a\|^2 + 2\,\|b\|^2, \\ &3) \quad \frac{1}{2}\,\|a\|^2 - \|b\|^2 \leq \|a+b\|^2. \end{aligned}$$

Proof of 1)

Let $a' = \frac{a}{\sqrt{t}}$, $b' = b\sqrt{t}$, i.e., $a = a'\sqrt{t}$, $b = \frac{b'}{\sqrt{t}}$. Then

$$\langle a, b\rangle = \langle a', b'\rangle, \qquad \frac{\|a\|^2}{2t} = \frac{\|a'\|^2}{2}, \qquad \frac{t\,\|b\|^2}{2} = \frac{\|b'\|^2}{2}.$$

As a result, we only have to prove

$$2\,\langle a', b'\rangle \leq \|a'\|^2 + \|b'\|^2,$$

which is equivalent to $\|a' - b'\|^2 \geq 0$, so it is true.

Proof of 2)

Using $\|a-b\|^2 \geq 0$ we get

$$\|a\|^2 + \|b\|^2 - 2\,\langle a, b\rangle \geq 0.$$

Adding $\|a\|^2 + \|b\|^2$ to both sides:

$$2\,\|a\|^2 + 2\,\|b\|^2 - 2\,\langle a, b\rangle \geq \|a\|^2 + \|b\|^2.$$

Adding $2\,\langle a, b\rangle$ and regrouping:

$$2\,\|a\|^2 + 2\,\|b\|^2 \geq \|a\|^2 + 2\,\langle a, b\rangle + \|b\|^2 = \|a+b\|^2.$$

Proof of 3)

3) is the same as

$$\|a\|^2 \leq 2\,\|a+b\|^2 + 2\,\|b\|^2.$$

Setting $a := a' + b'$, $b := -b'$ (so that $a + b = a'$) we get

$$\|a' + b'\|^2 \leq 2\,\|a'\|^2 + 2\,\|b'\|^2,$$

which is the same as 2).

Exercise 25

For a random vector $X \in \mathbb{R}^d$ (with finite second moment) and any $c \in \mathbb{R}^d$,

$$\mathbb{E}\big[\|X - \mathbb{E}[X]\|^2\big] = \mathbb{E}\big[\|X - c\|^2\big] - \|\mathbb{E}[X] - c\|^2.$$

Proof

$$\begin{aligned}\mathbb{E}\big[\|X-\mathbb{E}[X]\|^2\big] &= \mathbb{E}\big[\|(X-c)-(\mathbb{E}[X]-c)\|^2\big]\\ &= \mathbb{E}\big[\|X-c\|^2\big]+\mathbb{E}\big[\|\mathbb{E}[X]-c\|^2\big]-2\,\mathbb{E}\big[\langle X-c,\mathbb{E}[X]-c\rangle\big].\end{aligned}$$

On the other hand, $\|\mathbb{E}[X]-c\|^2$ is a number, so its expectation is the same number:

$$\mathbb{E}\big[\|\mathbb{E}[X]-c\|^2\big]=\|\mathbb{E}[X]-c\|^2.$$

So we get

$$\mathbb{E}\big[\|X-\mathbb{E}[X]\|^2\big]=\mathbb{E}\big[\|X-c\|^2\big]-\|\mathbb{E}[X]-c\|^2-2\Big(\mathbb{E}\big[\langle X-c,\mathbb{E}[X]-c\rangle\big]-\|\mathbb{E}[X]-c\|^2\Big).$$

Also

$$\begin{aligned}\mathbb{E}\big[\langle X-c,\mathbb{E}[X]-c\rangle\big] &= \mathbb{E}\big[\langle X,\mathbb{E}[X]\rangle\big]-\langle c,\mathbb{E}[X]\rangle-\langle c,\mathbb{E}[X]\rangle+\|c\|^2\\ &= \|\mathbb{E}[X]\|^2-2\,\langle c,\mathbb{E}[X]\rangle+\|c\|^2=\|\mathbb{E}[X]-c\|^2,\end{aligned}$$

which means

$$-2\Big(\mathbb{E}\big[\langle X-c,\mathbb{E}[X]-c\rangle\big]-\|\mathbb{E}[X]-c\|^2\Big)=0,$$

which proves the identity.

Exercise 26

Prove that for a random vector $X\in\mathbb{R}^d$ and any $t>0$ we have

$$\mathrm{Prob}\big(\|X-\mathbb{E}[X]\|\geq t\big)\leq\frac{\mathrm{Var}[X]}{t^2}.$$

Below $\mu:=\mathbb{E}[X]$ and $\sigma^2:=\mathrm{Var}[X]=\mathbb{E}\big[\|X-\mu\|^2\big]$.

Proof 1 (from scratch)

Replacing t by $t\sigma$ (for $\sigma>0$),

$$\mathrm{Prob}\big(\|X-\mu\|\geq t\big)\leq\frac{\sigma^2}{t^2}\iff\mathrm{Prob}\big(\|X-\mu\|\geq t\sigma\big)\leq\frac{1}{t^2}.$$

By the law of total expectation,

$$\begin{aligned}\sigma^2=\mathbb{E}\big[\|X-\mu\|^2\big] &= \mathbb{E}\big[\|X-\mu\|^2\;\big|\;\|X-\mu\|\geq t\sigma\big]\,\mathrm{Prob}\big(\|X-\mu\|\geq t\sigma\big)\\ &\quad+\mathbb{E}\big[\|X-\mu\|^2\;\big|\;\|X-\mu\|<t\sigma\big]\,\mathrm{Prob}\big(\|X-\mu\|<t\sigma\big)\\ &\geq (t\sigma)^2\,\mathrm{Prob}\big(\|X-\mu\|\geq t\sigma\big)+0\cdot\mathrm{Prob}\big(\|X-\mu\|<t\sigma\big)\\ &= (t\sigma)^2\,\mathrm{Prob}\big(\|X-\mu\|\geq t\sigma\big).\end{aligned}$$

As a result, $(t\sigma)^2\,\mathrm{Prob}\big(\|X-\mu\|\geq t\sigma\big)\leq\sigma^2$, which is the same as

$$\mathrm{Prob}\big(\|X-\mu\|\geq t\sigma\big)\leq\frac{1}{t^2}.$$

Proof 2 (using Markov's inequality)

Markov's inequality: for a non-negative random variable Z and $s > 0$,

$$\text{Prob}(Z \geq s) \leq \frac{\mathbb{E}[Z]}{s}.$$

Let $Z := \|X - \mu\|^2$, where $\mu = \mathbb{E}[X]$. Then

$$\text{Prob}\big(\|X - \mu\|^2 \geq s\big) \leq \frac{\mathbb{E}\big[\|X - \mu\|^2\big]}{s},$$

which is the same as

$$\text{Prob}\big(\|X - \mu\| \geq \sqrt{s}\big) \leq \frac{\text{Var}[X]}{s}.$$

Substituting $s = t^2$ we get

$$\text{Prob}\big(\|X - \mu\| \geq t\big) \leq \frac{\text{Var}[X]}{t^2},$$

which proves the inequality.